

Aejandro De La Torre

LIS 440

wk12: Pandas Subsetting and Slicing - Ice Cream Data Randomized

12/1/25

Code

```
# Import necessary packages
import pandas as pd
from pathlib import Path

# Print pandas version for tracking
print(pd.__version__)

# Personal tag
print("Alejandro De La Torre's Code...")

# -----
# File & Path Setup
# -----

# Set root directory (two levels up from this script)
root = Path(__file__).parent.parent.resolve()

# Show current working directory
print("Current Working Directory:", Path.cwd())

# Define key directories
data = root / 'data'
data_raw = data / 'raw'
data_processed = data / 'processed'
output = root / 'output'
scripts = root / 'scripts'

# Define path to CSV file
icecream_csv = data_processed / 'icecream_random.csv'

# Load CSV into DataFrame
ic_df = pd.read_csv(icecream_csv)

# Preview data
print("First 3 rows:\n", ic_df.head(3))
print("Last 3 rows:\n", ic_df.tail(3))

# -----
# Total scoops across all containers
# -----

# Calculate total scoops per flavor across all columns except 'Flavor'
ic_df['Total Scoops (All Containers)'] = ic_df.iloc[:, 1:].sum(axis=1)

# Display results
print("\nTotal Scoops for Each Flavor Across All Containers:")
print(ic_df[['Flavor', 'Total Scoops (All Containers)']])
```

```
# -----
# Total scoops for selected containers only
# -----

# Define selected container columns
selected_containers = ['Waffle Cone', 'Cake Cone', 'Pretzel Cone']

# Calculate total scoops per flavor for selected containers
ic_df['Total Scoops (Selected Containers)'] = ic_df[selected_containers].sum(axis=1)

# Display results
print("\nTotal Scoops for Each Flavor for Waffle Cone, Cake Cone, and Pretzel Cone:")
print(ic_df[['Flavor', 'Total Scoops (Selected Containers)']])
```

Shell

```
>>> %Run random_icecream.py
2.3.3
Alejandro De La Torre's Code...
Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S
440/Lectures/Week 13/icecream_random/scripts
First 3 rows:
   Flavor Sugar Cone Cup Waffle Cone Cake Cone Pretzel Cone
0  Chocolate      25  32      26    25      26
1  Strawberry      40  42      45    29      39
2   Vanilla      43  41      30    41      35
Last 3 rows:
   Flavor Sugar Cone Cup Waffle Cone Cake Cone Pretzel Cone
7  Black Cherry      43  40      43    38      26
8   Orange      29  31      39    26      34
9  Cookie Dough      34  38      35    32      45

Total Scoops for Each Flavor Across All Containers:
   Flavor Total Scoops (All Containers)
0  Chocolate              134
1  Strawberry              195
2   Vanilla              190
3  Moose Tracks           183
4   Lemon              167
5  Neapolitan            178
6  Blue Moon            184
7  Black Cherry           190
8   Orange              159
9  Cookie Dough           184

Total Scoops for Each Flavor for Waffle Cone, Cake Cone, and Pretzel Cone:
   Flavor Total Scoops (Selected Containers)
0  Chocolate              77
1  Strawberry             113
2   Vanilla             106
3  Moose Tracks           105
4   Lemon              106
```

5	Neapolitan	104
6	Blue Moon	110
7	Black Cherry	107
8	Orange	99
9	Cookie Dough	112
>>>		